(ii) The probability that the student participated in neither of the two tournaments in

$$P(\overline{A} \cap \overline{B}) = P(\overline{A \cup B}) \text{ (By De Morgan's law } \overline{A \cup B} = \overline{A} \cap \overline{B})$$

$$= 1 - P(A \cup B)$$

$$= 1 - \frac{8}{13} = \frac{5}{13}$$

(iii) The probability that the student participated in Hockey only is

$$P(\overline{A} \cap B) = P(B) - P(A \cap B)$$

$$= \frac{120}{260} - \frac{50}{260} = \frac{70}{260} = \frac{7}{26}$$

(iv) The probability that the student participated in Kabadi only

$$P(A \cap \overline{B}) = P(A) - P(A \cap B)$$

$$= \frac{90}{260} - \frac{50}{260} = \frac{40}{260} = \frac{2}{13}$$

(v) The probability that the student participated in exactly one of the tournaments is

$P[(A \cap \overline{B}) \cup (\overline{A} \cap B)] = P(A \cap \overline{B}) + P(\overline{A} \cap B)$ [$\because A \cap \overline{B}, \overline{A} \cap B$ are mutually exclusive events]

$$= \frac{70}{260} + \frac{40}{260} = \frac{110}{260} = \frac{11}{26}$$

8.5 Conditional Probability

Consider the following situations:

(i) two events occur successively or one after the other (e.g) A occurs after B has occurred and

(ii) both event A and event B occur together.

Example 8.13

There are 4000 people living in a village including 1500 female. Among the people in the village, the age of 1000 people is above 25 years which includes 400 female. Suppose a person is chosen and you are told that the chosen person is a female. What is the probability that her age is above 25 years?